Latent Diffusion for Brain-Tumour MRI Augmentation: Memorisation, the Autoencoder Bottleneck, and What Still Helps

[First author (project lead), affiliation] Syed Muhammad Hamza Shah*

Draft of September 11, 2026

Abstract

Latent diffusion models (LDMs) built on a pretrained natural-image autoencoder are the standard recipe for synthetic medical images, but whether their samples help a downstream task is rarely tested against the confounds that decide the answer. We train mask-conditioned 2-D LDMs on the BraTS 2020 training set with patient-level splits and evaluate them with a protocol fixed before the results were seen: early stopping on validation loss, a nearest-neighbour memorisation check, model selection on validation data only, and a segmentation study with three seeds, patient-matched synthetic pairs and pre-declared secondary analyses. Three findings follow. (1) Without regularisation the LDM memorises: by epoch 200 its FID keeps improving while 97 % of its samples lie closer to a training slice than real held-out slices do; cached affine augmentation of the latents postpones this and yields the best early-stopped model (FID 24.7 vs 29.9 on validation). (2) Mixing synthetic slices into segmenter training *lowers* tumour-core and enhancing-tumour Dice at 128 px at every data fraction (−0.030 to −0.036 mean Dice); real slices merely passed through the frozen autoencoder lose even more (−0.057 to −0.072), and fine-tuning the autoencoder's decoder on the training slices, which raises the reconstruction ceiling from 26.4 to 27.8 dB and cuts per-modality FIDs by 35–75 %, recovers only about a third of the loss. The bottleneck is the encoder's $16 \times 16 \times 4$ latent, not the diffusion model. (3) Two uses of the same samples do help: pre-training the segmenter on synthetic pairs before fine-tuning on real data gains +0.021 / +0.014 mean Dice at 10 % / 25 % real data over a compute-matched control, and at 256 px, where the generator reaches the FID floor between two real sets (10.45 vs 9.55), mixing gains +0.027 with 26 real patients. Code, splits and run logs are released.

1 Introduction

Synthetic images are an attractive answer to the scarcity of labelled medical data, and denoising diffusion models [14, 24] now produce brain MRI slices that are hard to tell from real ones [17, 18, 23]. The usual evidence for their usefulness is a fidelity score such as FID [12] and, sometimes, a segmenter trained with the synthetic images added. Both are easy to get wrong. Fidelity scores improve when a model copies its training set [1, 6, 8, 26], so the checkpoint with the best FID may be the one that has memorised the most. And a downstream gain measured with one seed, slice-level splits, or synthetic images whose conditioning masks come from held-out patients can be an artefact of the evaluation rather than of the data.

This paper asks a narrow question carefully: does a mask-conditioned latent diffusion model trained on BraTS 2020 produce (image, mask) pairs that improve a 2-D tumour segmenter, and if not, why not? We fix the evaluation before looking at the results: patient-level splits, early stopping on the validation diffusion loss, a flip-aware nearest-neighbour memorisation check, model selection on validation data only, three seeds per segmentation condition, patient-matched synthetic pairs, and secondary analyses declared in the repository before they were run. Our contributions are:

- A checkpoint-curve analysis showing that the unregularised LDM's FID keeps improving after its validation loss turns, but only by memorising (Sect. 5.1); cached affine augmentation of the latents postpones memorisation and gives the best early-stopped model, whereas dropout alone changes nothing.
- A negative downstream result with its cause isolated (Sect. 5.3–5.6): at 128 px, mixing synthetic pairs into training lowers tumour-core (TC) and enhancing-tumour (ET) Dice at every data fraction. Real slices passed through the frozen Stable Diffusion autoencoder lose even more, and a decoder fine-tuned on the training slices, although it improves every image metric, recovers only a third of the Dice. The loss is the encoder's latent bottleneck, not the diffusion model's.

*[Affiliation and contact e-mail.]

- Two positive results that survive their controls (Sect. 5.4, 5.5): synthetic pre-training helps in the low-data regime even against a control given the same extra epochs on real data, and at 256 px, where the generator sits at the FID floor, mixing helps with 26 real patients.

2 Related work

Diffusion models for medical image synthesis. Pinaya et al. [23] trained a 3-D LDM on 31,740 UK Biobank brains; Khader et al. [18] and Müller-Franzes et al. [22] compared diffusion models with GANs on several modalities and found diffusion samples more realistic. Fernandez et al. [9] trained segmenters on fully synthetic brain data and reported near-parity with real data for some tasks. Surveys [7, 17] list many more. Most of this work reports fidelity metrics or a single downstream run; few report a memorisation check or a control for the autoencoder.

Memorisation. Carlini et al. [6] and Somepalli et al. [26] showed that diffusion models reproduce training images, and Gu et al. [11] related the effect to data size and training length. In medical imaging, Dar et al. [8] found that a 3-D LDM memorised a large share of its training volumes, and Akbar et al. [1] reported the same for 2-D brain MRI and chest X-rays, both with nearest-neighbour distances as we do. Data augmentation is the classical remedy for generator overfitting [16]; we apply it in latent space so that the frozen autoencoder is run once per variant.

Synthetic data for segmentation. GAN-based augmentation helped classification in early work [10], but gains for segmentation are mixed and depend on the ratio of synthetic to real data and on how the synthetic labels are obtained. We use the conditioning mask as the label, restrict each low-data segmenter to synthetic slices conditioned on masks of its own patients, and test three uses of the same pool (mixing, a 1:1 ratio, pre-training).

3 Data and protocol

Data. The BraTS 2020 training set [2, 3, 21] contains 369 patients with co-registered, skull-stripped FLAIR, T1, T1ce and T2 volumes ($240 \times 240 \times 155$ at $1\,\mathrm{mm}^3$) and expert labels (necrotic core, oedema, enhancing tumour). Patients are split once, at patient level, into 258 training, 37 validation and 74 test patients (seed 0; the split file is committed). Each modality of each volume is clipped to its [0.5, 99.5] intensity percentiles inside the brain and rescaled to $[0, 1]$; axial slices whose tumour covers at least 0.1 % of the slice are kept, centre-cropped to 224×224 (the brain never extends beyond that window) and resized to 128 or 256 px. This gives 15,895 / 2,159 / 4,623 tumour-bearing slices for train / val / test. FLAIR, T1ce and T2 are mapped to the three input channels of the pretrained autoencoder. Evaluation regions follow the challenge: whole tumour (WT), tumour core (TC) and enhancing tumour (ET).

Pre-declared protocol. Every analysis below was written into the repository's protocol file before its results were seen; the timestamps and the exact reading rules are in the released `docs/EXPERIMENTS.md`. In particular the checkpoint rule, the model-selection rule, the segmentation protocol, its three secondary analyses, the compute-matched control, the 256 px repeat and the decoder fine-tune study (with the condition under which it would stop early) were fixed in advance. All numbers come from run directories that record the resolved configuration, the git commit and the metrics.

4 Methods

4.1 Latent diffusion model

We follow Rombach et al. [24]. The frozen Stable Diffusion autoencoder `sd-vae-ft-mse` compresses a slice by $8\times$ into a 4-channel latent ($16 \times 16 \times 4$ at 128 px, $32 \times 32 \times 4$ at 256 px). A U-Net (channels 128/256/512/512, two residual blocks per level, self-attention at the two coarsest levels, 102 M parameters) predicts the noise ϵ under a 1,000-step linear schedule [14]. Training uses AdamW [19] (lr 10^{-4}, weight decay 0.01), 500 warm-up steps then cosine decay, gradient clipping at 1.0, an EMA of the weights (0.9999) that is used for every reported sample, and bf16 autocast. Batch size is 64 at 128 px and 32 at 256 px; 200 epochs correspond to 49.6k and 99.2k steps and take 1.5 h and 2.8 h on one RTX A6000.

Mask conditioning. The four-class mask is one-hot encoded, average-pooled to the latent grid (soft per-cell class fractions) and concatenated to the noisy latent, together with a fifth *null* channel that marks a dropped condition so that "no condition" is distinct from "all background". Conditions are dropped with probability 0.1 during training, which enables classifier-free guidance [13] at sampling time. Samples are drawn with DDIM [27], 50 steps, $\eta = 0$, guidance scale 2 unless stated.

Cached latents and augmentation. Because the autoencoder is frozen, the posterior of every slice is encoded once and cached (mean and log-variance, sampled on each access, for the slice and its horizontal flip). The regularised recipe ("reg") adds U-Net dropout 0.1 and six extra cached variants per training slice, each a random affine transform (flip with $p = 0.5$, shift up to 6 %, rotation up to $\pm 10°$, isotropic scale 0.9–1.1; zero padding). Each access draws one of the eight variants and applies the identical transform to the conditioning mask. Validation latents are never augmented, so validation losses remain comparable across recipes.

4.2 Checkpoint selection and memorisation check

The validation diffusion loss is computed with fixed noise and evenly spaced timesteps so that epochs are comparable; the EMA weights of the epoch with the lowest validation loss are the reported model. To see what later checkpoints buy, every tenth epoch is scored with the same 2,000 masks and seeds: FID/KID [4, 12] against the real validation slices and a memorisation statistic. For the latter, each sample is compared with every training slice *in both orientations* (the models train with flips, so a copy of a mirrored slice is a copy) by L_2 distance on 64×64 downsampled images; the same distances are computed for real held-out slices, and the *memorised fraction* is the share of samples closer to a training slice than 95 % of real held-out slices are. A model that generalises scores about 0.05 by construction.

Model selection. Three 128 px mask-conditioned candidates differ only in regularisation: flips only (baseline), plus dropout 0.1, and the reg recipe. The rule, fixed before the runs finished, takes each candidate at its best-validation-loss checkpoint, excludes candidates with memorised fraction above 0.15, and picks the lowest FID against the validation slices. The test set plays no part. The chosen recipe was then reused unchanged for an unconditional 128 px model and for the 256 px model.

4.3 Generation metrics

Against all 4,623 real test slices we report FID and KID (InceptionV3 features via torch-fidelity) for the three-channel image and for each modality with the grey channel replicated to RGB, using 5,000 samples per setting; the FID between the real validation and real test slices (10.49 at 128 px, 9.55 at 256 px) is the floor these numbers should be read against. Diversity is the mean SSIM [28] over 2,000 random sample pairs (real test pairs: 0.652 at 128 px, 0.690 at 256 px). The autoencoder's *ceiling*, PSNR / SSIM / LPIPS [29] of decode(encode(x)) on real test slices, bounds every latent model. Each mask-conditioned model also generates 5,000 samples from masks of *validation* patients, tumour shapes it never saw.

4.4 Downstream segmentation

A 2-D U-Net [25] from MONAI [5] (channels 32–512, two residual units per level, instance normalisation) is trained with Dice + cross-entropy, AdamW (3×10^{-4}, weight decay 10^{-4}) with a one-cycle schedule, 40 epochs, batch 32 and horizontal flips; the epoch with the best validation Dice on real validation patients is kept and scored on the 74 real test patients. Dice is computed per patient over that patient's slices and then averaged over patients; we report WT, TC, ET and their mean, as mean ± standard deviation over three seeds. Conditions: 10 % (26 patients), 25 % (64) and 100 % (258) of the training patients, each with and without synthetic pairs, plus synthetic-only. The pool is the selected model's 20,000-sample set conditioned on training masks, each sample paired with its mask. *Patient matching*: a real-x % segmenter only receives synthetic slices whose mask belongs to one of its own x % patients, so a low-data condition never sees tumour geometry of patients it does not have (about 1.25 synthetic slices per real slice). The real-only segmenters are also scored on all synthetic pairs ("mask consistency": Dice between the prediction on the generated image and the conditioning mask).

Secondary analyses (declared after the first seed-0 results). (i) a 1:1 synthetic ratio; (ii) synthetic pre-training: 20 epochs on the synthetic pairs, then the standard 40 epochs on real data; (iii) VAE-reconstructed real: the real slices are replaced by their own reconstruction through the frozen autoencoder, with no generator involved, which measures the autoencoder's share of any loss. A compute-matched control for (ii), declared after its first result, pre-trains for the same 20 epochs on the real slices themselves.

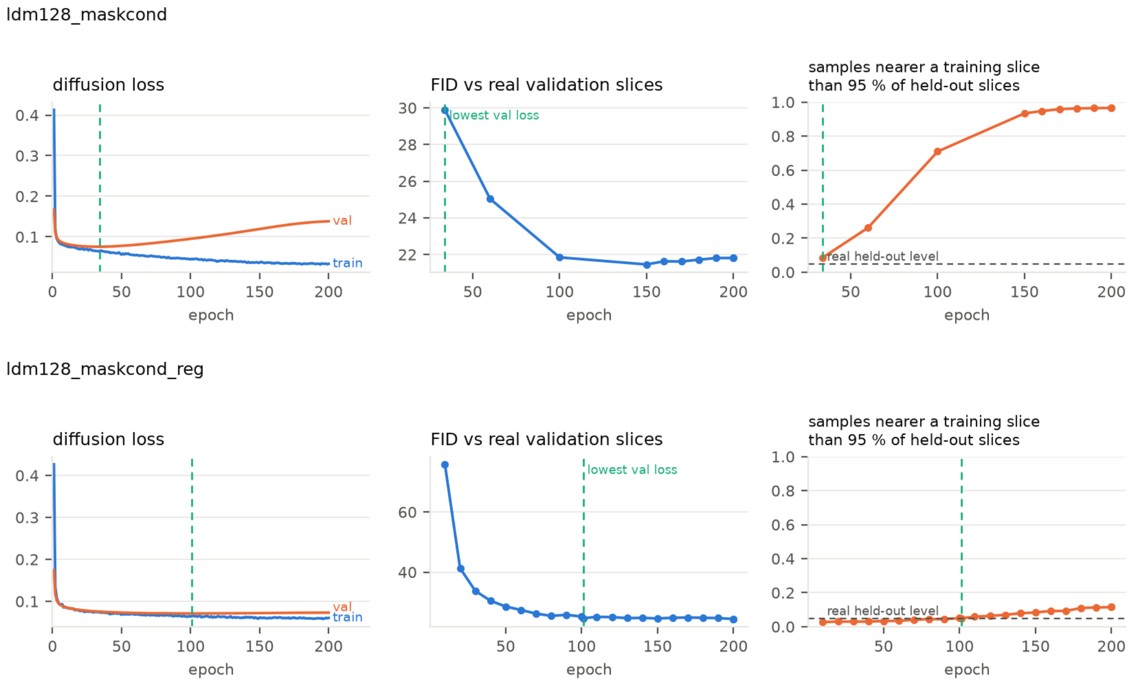

Figure 1: Validation diffusion loss, FID against real validation slices and memorised fraction per saved checkpoint for the unregularised 128 px mask-conditioned model (top) and the regularised recipe (bottom). FID keeps improving after the validation-loss minimum in both, but the unregularised model does so by copying: 97 % of samples are closer to a training slice than real held-out slices are by epoch 200.

Table 1: Checkpoint curves (2,000 samples per checkpoint, FID vs real validation slices). "Best" is the epoch with the lowest validation diffusion loss, the checkpoint used everywhere else. The selection rule (validation data only) picked the reg recipe.

model	best epoch	$\mathrm{FID}_{\mathrm{val}}$	$\mathrm{KID}\times 10^3$	memorised	$\mathrm{FID}_{\mathrm{val}}$ @200	memorised @200
LDM-128-mask (flips only)	34	29.89	18.18	0.086	21.81	**0.966**
+ dropout 0.1	41	28.82	17.24	0.093	21.99	**0.891**
+ dropout 0.1 + affine augmentation (reg, chosen)	101	**24.65**	**12.86**	**0.049**	24.43	0.115
LDM-128-reg (unconditional)	116	24.81	15.87†	0.040	23.20	0.051
LDM-256-mask-reg	71	15.10	–	0.033	12.94	0.291

†KID of the unconditional model is against the test slices (Table 2); the curve records FID only.

4.5 Decoder fine-tuning

To test whether the autoencoder's share is the decoder's blur, we fine-tune only the decoder and its post-quantisation convolution (49.5 M parameters) on the 15,895 training slices with $L_1 + 0.5{\cdot}$LPIPS-VGG per channel (AdamW 2×10^{-5}, batch 16, flips, 8 epochs, 57 min); the epoch with the lowest validation loss is kept. The encoder is untouched, so every cached latent and every trained diffusion model stay valid: the *same* latent samples are decoded again with the new decoder and re-scored, and the primary and secondary segmentation protocols are repeated with that pool, the VAE-reconstructed-real control now using the fine-tuned decoder. Two readings were fixed in advance: (a) the study stops if the reconstruction ceiling does not improve; (b) the loss is attributed to the decoder only if the VAE-reconstructed-real control moves towards real-only *and* real + synthetic moves the same way.

5 Results

5.1 Memorisation versus training length

Figure 1 and Table 1 give the answer to "how long should one train?". The baseline reaches its lowest validation loss at epoch 34; by epoch 200 its FID against real validation slices has fallen from 29.9 to 21.8, an improvement any FID-based selection would take, but 97 % of its samples then lie closer to a training slice than 95 % of real held-out slices do. Dropout alone changes nothing (0.891 memorised at epoch 200). Cached affine augmentation postpones the turn to epoch 101, lowers the memorised fraction at the last epoch to

Table 2: Generation quality against all 4,623 real test slices (5,000 samples; best-validation-loss EMA weights; DDIM 50). The FID between real validation and real test slices is 10.49 (128 px) and 9.55 (256 px); real pair-SSIM is 0.652 / 0.690. Per-modality FIDs replicate one grey channel to RGB and are far above the composite.

model	setting	FID	KID$\times10^3$	FID FLAIR / T1ce / T2	pair-SSIM	memorised
LDM-128-mask (baseline)	g=2	26.22	20.89	45.9 / 45.4 / 73.0	0.576	0.069
LDM-128-mask + dropout	g=2	24.34	19.13	47.7 / 46.2 / 75.7	0.585	0.070
LDM-128-mask-reg	g=2	20.85	15.19	44.3 / 44.0 / 69.6	0.584	0.043
LDM-128-mask-reg	g=1	20.37	14.74	47.0 / 44.2 / 69.4	0.586	0.036
LDM-128-mask-reg	g=2, unseen masks	22.85	16.52	45.2 / 44.1 / 71.1	0.593	0.040
LDM-128-reg (unconditional)	g=1	21.31	15.87	47.5 / 45.0 / 68.7	0.579	0.027
LDM-256-mask-reg	g=2	**10.45**	**6.40**	40.3 / 39.6 / 59.1	0.628	0.030
LDM-256-mask-reg	g=1	12.83	8.77	43.1 / 41.7 / 58.2	0.637	0.021
LDM-256-mask-reg	g=2, unseen masks	10.83	6.17	40.4 / 39.3 / 60.7	0.633	0.024

Table 3: Per-patient test Dice at 128 px (74 test patients; mean ± std over 3 seeds). Synthetic pairs come from LDM-128-mask-reg and are patient-matched.

training data	patients	real slices	synthetic	WT	TC	ET	mean
10 % real	26	1,483–1,705	0	0.801 ± 0.011	0.580 ± 0.023	0.509 ± 0.022	0.630 ± 0.016
10 % real + synthetic	26	1,483–1,705	1,845–2,208	0.809 ± 0.006	0.559 ± 0.010	0.430 ± 0.033	0.600 ± 0.014
25 % real	64	3,937–4,031	0	0.847 ± 0.003	0.676 ± 0.011	0.611 ± 0.010	0.711 ± 0.004
25 % real + synthetic	64	3,937–4,031	4,913–5,158	0.844 ± 0.004	0.660 ± 0.021	0.521 ± 0.014	0.675 ± 0.012
100 % real	258	15,895	0	0.886 ± 0.001	0.775 ± 0.005	0.673 ± 0.004	0.778 ± 0.003
100 % real + synthetic	258	15,895	20,000	0.879 ± 0.001	0.742 ± 0.009	0.650 ± 0.010	0.757 ± 0.006
synthetic only	0	0	20,000	0.829 ± 0.002	0.669 ± 0.007	0.481 ± 0.012	0.660 ± 0.005

0.115 and gives the best early-stopped model (FID 24.7 vs 28.8 and 29.9). Even so, the regularised 256 px model starts copying after epoch ≈ 90 (0.074 at epoch 100, 0.291 at epoch 200): augmentation postpones memorisation, it does not remove it, and early stopping on validation loss remains necessary. All results below use the best-validation-loss checkpoint.

5.2 Generation quality on the test set

Table 2 scores the early-stopped models on the held-out patients. The regularised recipe improves test FID by 5.4 points over the baseline at 128 px. Masks of validation patients, tumour shapes never seen in training, cost 2.0 FID points at 128 px and 0.4 at 256 px, so the models generalise to new tumour geometry rather than redrawing training tumours. Conditioning costs no fidelity (the mask-conditioned and unconditional 128 px models score alike), guidance 2 helps at 256 px and is marginally worse than guidance 1 at 128 px, and every memorised fraction is at or below the 0.05 expected of a model that generalises. At 256 px the model reaches FID 10.45 against a floor of 9.55 between two sets of *real* slices (Fig. 2). T2 is the hardest modality at both resolutions. The autoencoder ceiling is PSNR 26.4 dB / SSIM 0.833 / LPIPS 0.041 at 128 px and 29.0 dB / 0.877 / 0.036 at 256 px; no latent model can be sharper than this.

5.3 Downstream segmentation at 128 px

Mixing the synthetic pairs into training did not improve the segmenter (Table 3, Fig. 3). Whole-tumour Dice rises slightly with 26 real patients (0.801 → 0.809), but TC and ET fall at every fraction (ET at 10 %: 0.509 → 0.430; mean Dice at 100 %: 0.778 → 0.757), and the loss is present in 9 of 9 seed-matched pairs. Synthetic data alone reaches 0.660 mean Dice, about the level of 26 real patients. Mask consistency explains where the loss sits: a real-trained segmenter recovers the whole-tumour outline of the synthetic images well (WT consistency 0.789 for the 100 % segmenter, vs 0.886 on real test slices) but agrees with the conditioning mask on ET only about half the time (0.494). The pairs carry the tumour outline but a blurred TC/ET appearance, and a segmenter trained on them learns that appearance.

5.4 Secondary analyses: why mixing hurts, and what helps instead

Table 4 and Fig. 4 answer the three questions declared after the first seed-0 results. *The ratio is not the cause*: capping synthetic slices at 1:1 recovers only about a third of the loss and ET stays 0.03–0.07 below real-only. *The frozen autoencoder alone reproduces the loss, and more*: replacing the real slices by their own reconstruction through `sd-vae-ft-mse`, with no generator involved, costs −0.057 / −0.072 / −0.072

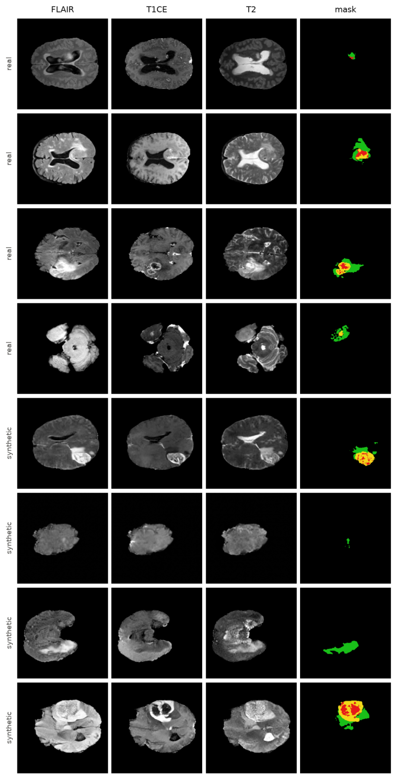

Figure 2: Real test slices (top four rows) and LDM-256-mask-reg samples (bottom four rows) with their masks. Colours: necrotic core, oedema, enhancing tumour.

Table 4: Secondary analyses at 128 px: change in mean test Dice relative to real only, seed-matched, three seeds (per-region changes for the two decisive rows). The control row pre-trains for the same 20 extra epochs on the real slices.

condition	10 % real	25 % real	100 % real
real + synthetic, 1.25:1 (primary)	−0.030	−0.036	−0.021
real + synthetic, 1:1	−0.020	−0.024	−0.018
real replaced by its frozen-VAE reconstruction (no generator)	−0.057 (ET −0.112)	−0.072 (ET −0.135)	−0.072 (ET −0.131)
synthetic pre-training, then real	**+0.044**	**+0.015**	+0.001
real pre-training, then real (compute-matched control)	+0.023	+0.001	+0.004
synthetic pre-training − control	**+0.021** (3/3 seeds)	**+0.014** (3/3 seeds)	−0.003

mean Dice, again concentrated on ET (−0.11 to −0.14). Every synthetic image passes through exactly this autoencoder, so the enhancing-tumour detail the segmenter needs is removed before the diffusion model is involved. *Pre-training helps where mixing does not*: 20 epochs on synthetic pairs followed by the standard schedule on real data gains +0.044 mean Dice with 26 patients (all three regions, ET 0.509 → 0.564, 3/3 seeds) and +0.015 with 64, nothing with 258. Because the pre-trained segmenter sees 20 extra epochs, a control was declared that spends those epochs on the real slices: it recovers about half of the 10 % gain and none of the 25 % gain, leaving a synthetic-specific effect of +0.021 and +0.014 (3/3 seeds each). Fine-tuning on real slices overrides the blurred appearance that mixing bakes in, while the features learned from the synthetic pairs still transfer.

5.5 Segmentation at 256 px

The primary protocol was repeated unchanged at 256 px, declared before any 256 px segmenter was trained (Table 5). With 26 real patients the sign flips: the patient-matched synthetic pairs improve every region (TC +0.049, ET +0.022, WT +0.011; 3/3 seeds, spread $\leq$ 0.004), where the same protocol at 128 px lost 0.030. With 64 patients the effect is neutral and with 258 it is neutral on the mean and still slightly negative on ET (−0.015). The synthetic-only segmenter reaches 0.760, the level of 25 % real data (128 px: 0.660, the level of 10 %). The 256 px pool comes from a generator at the FID floor decoded through an autoencoder whose ceiling is 2.6 dB higher; the real-only segmenters are stronger at 256 px as well, so the two resolutions are

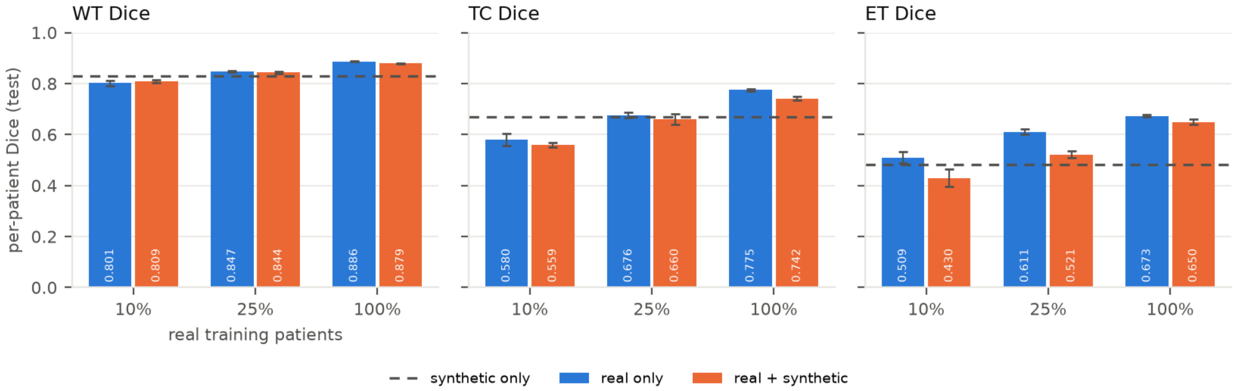

Figure 3: 128 px primary protocol: per-patient test Dice with and without patient-matched synthetic pairs, three seeds. The dashed line is the synthetic-only segmenter.

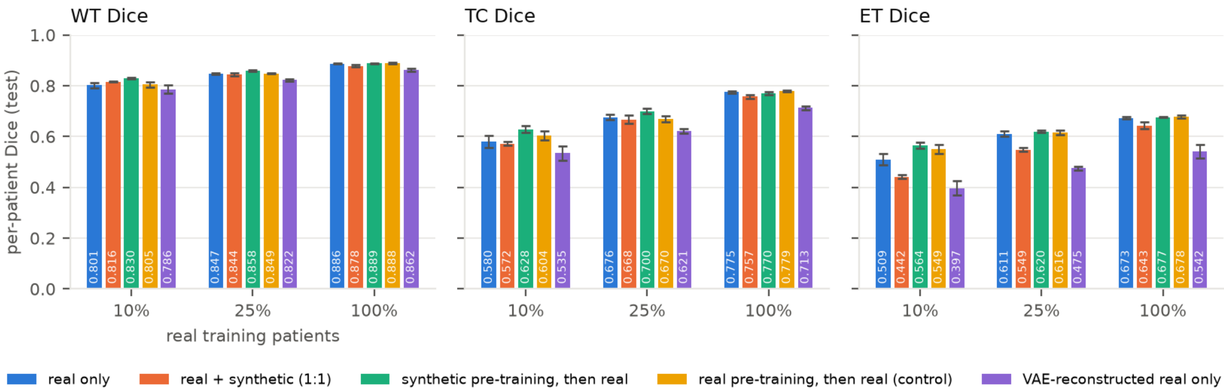

Figure 4: 128 px secondary analyses (frozen decoder): real only, real + synthetic 1:1, synthetic pre-training, real pre-training (control) and real slices reconstructed through the frozen autoencoder.

compared only within themselves.

5.6 Is it the decoder? Fine-tuning the autoencoder's decoder

Fine-tuning the decoder on the training slices improves every image metric (Table 6, Fig. 5): the ceiling rises from 26.37 to 27.79 dB (SSIM 0.833 → 0.883, LPIPS 0.041 → 0.032; T2, the worst channel, gains 1.6 dB), so reading (a) passed and the study continued. Decoding the same 20,000 latents again leaves the three-channel FID unchanged (20.85 → 20.89) but cuts the per-modality FIDs by 35–75 % (T2 69.6 → 17.6); the per-channel scores of the frozen decoder were dominated by the texture it invents, which the composite metric, seeing the three channels as one colour image, largely ignores. Sample diversity moves towards that of real slices (pair-SSIM 0.584 → 0.636, real 0.652), and the memorised fraction rises from 0.043 to 0.051 because every sample moves closer to every real slice under a sharper decoding; the latents are identical, so this is not new copying.

The segmenter, however, barely notices. With the re-decoded pool, mixing still costs Dice at every fraction and in 9 of 9 seed-matched pairs (−0.020 / −0.033 / −0.012; ET still −0.06 to −0.08), the re-trained real-only baselines reproduce the earlier ones within the seed spread, and the synthetic-only segmenter improves from 0.660 to 0.672. The VAE-reconstructed-real control moves towards real-only at every fraction (−0.057 / −0.072 / −0.072 → −0.044 / −0.055 / −0.045) and real + synthetic moves the same way, so the pre-declared condition (b) is met, but only in part: a decoder trained on these very slices still loses 0.044–0.055 mean Dice (ET 0.08–0.11) when real slices are merely passed through the autoencoder, which remains more than mixing costs. The loss is therefore the autoencoder's and not the diffusion model's, and its larger part sits in the encoder's $16 \times 16 \times 4$ latent, which discards enhancing-tumour detail that no decoder can restore; the decoder accounts for about a third. The pre-training gain is unchanged by the decoder (+0.018 / +0.015 against the control).

Table 5: Per-patient test Dice at 256 px with the LDM-256-mask-reg pool (same protocol, seeds and patient matching as Table 3).

training data	WT	TC	ET	mean	Δ mean
10 % real	0.849 ± 0.006	0.672 ± 0.019	0.604 ± 0.009	0.709 ± 0.003	
10 % real + synthetic	0.860 ± 0.006	0.721 ± 0.007	0.626 ± 0.006	0.736 ± 0.004	+**0.027**
25 % real	0.874 ± 0.003	0.751 ± 0.009	0.668 ± 0.010	0.764 ± 0.007	
25 % real + synthetic	0.878 ± 0.000	0.769 ± 0.004	0.654 ± 0.000	0.767 ± 0.001	+0.003
100 % real	0.895 ± 0.000	0.810 ± 0.002	0.707 ± 0.002	0.804 ± 0.000	
100 % real + synthetic	0.894 ± 0.004	0.818 ± 0.006	0.692 ± 0.008	0.801 ± 0.005	−0.003
synthetic only	0.871 ± 0.002	0.785 ± 0.008	0.623 ± 0.004	0.760 ± 0.003	

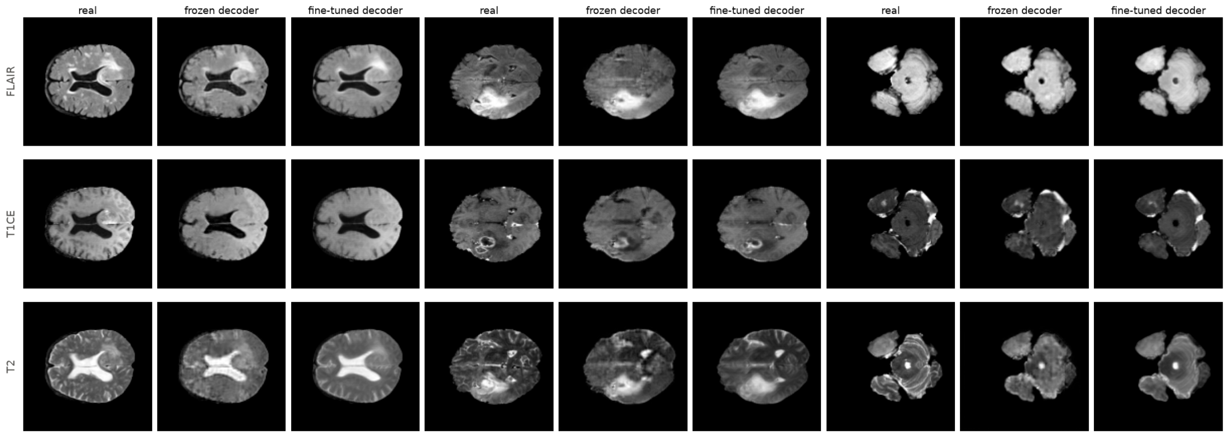

Figure 5: Three test slices with enhancing tumour reconstructed through the frozen and the fine-tuned decoder (same encoder, same latents). The blotchy texture the frozen decoder adds to T2 disappears and enhancing rims in T1ce come out sharper.

6 Discussion

What the results say. A latent diffusion model on a natural-image autoencoder can generate BraTS slices that score at the FID floor between real sets, generalise to unseen tumour shapes and, when trained long enough, copy its training set while every fidelity metric keeps improving. Whether its samples help a segmenter is decided by the autoencoder, not by the generator: at 128 px the latent bottleneck removes the enhancing-tumour detail that separates TC and ET, so mixing the samples into training teaches the segmenter a blurred appearance and lowers Dice, and a decoder fine-tuned on the target data improves every image metric without repairing this. Two uses survive their controls: pre-training on synthetic pairs and fine-tuning on real data, which lets the real data override the blurred appearance while keeping what transfers, and mixing at 256 px in the low-data regime, where the bottleneck is four times wider.

Implications for evaluation. Three practices were decisive and are cheap: early stopping on the validation diffusion loss with a memorisation check, since FID alone would have chosen the most memorised checkpoints; a no-generator control (real slices through the autoencoder), which localised the loss; and a compute-matched control for pre-training, which halved the apparent gain. Per-modality FIDs and the composite FID also disagree sharply once the decoder changes, so a single FID number is not a safe summary for multi-channel medical images.

Limitations. All models are 2-D and Dice is aggregated per patient over slices, not on volumes. FID/KID use Inception features; the real val-vs-test floor calibrates them, but a radiology-specific extractor [20] would be more sensitive to clinically relevant detail. The encoder of the autoencoder is frozen; a medical-image autoencoder or an encoder fine-tune with the diffusion models retrained is the natural next experiment, and this study predicts it should matter more than the decoder did. The segmenter is a plain 2-D U-Net rather than a tuned nnU-Net [15], and there is one dataset, one split and three seeds, with no external test cohort.

Table 6: Decoder fine-tuning at 128 px. Top: reconstruction ceiling on the 4,623 test slices and scores of the *same* 20,000 latent samples decoded with each decoder. Bottom: change in mean test Dice relative to real only (seed-matched, three seeds) when the segmentation protocols are repeated with the re-decoded pool.

	PSNR	SSIM	LPIPS	FID (3-ch)	FID FLAIR / T1ce / T2	memorised
frozen decoder	26.37 dB	0.833	0.041	20.85	44.3 / 44.0 / 69.6	0.043
fine-tuned decoder	**27.79 dB**	**0.883**	**0.032**	20.89	**28.6 / 25.6 / 17.6**	0.051

Δ mean Dice vs real only	10 % real		25 % real		100 % real	
	frozen	fine-tuned	frozen	fine-tuned	frozen	fine-tuned
real + synthetic (1.25:1)	−0.031	−0.020	−0.036	−0.033	−0.021	−0.012
ET only	−0.079	−0.064	−0.089	−0.081	−0.024	−0.022
real + synthetic (1:1)	−0.020	−0.007	−0.024	−0.025	−0.018	−0.012
real replaced by its VAE reconstruction	−0.057	−0.044	−0.072	−0.055	−0.072	−0.045
ET only	−0.112	−0.080	−0.135	−0.107	−0.131	−0.089
synthetic pre-training − control	+0.021	+0.018	+0.014	+0.015	−0.003	−0.001
synthetic only (mean Dice)	0.660 → 0.672					

7 Conclusion

On BraTS 2020, mask-conditioned latent diffusion with cached affine augmentation and early stopping yields a generator at the FID floor that does not memorise, but its 128 px samples do not help a tumour segmenter when mixed into training, because the frozen autoencoder's latent bottleneck, not the diffusion model and only in part its decoder, removes the enhancing-tumour detail the task depends on. The same samples do help as pre-training in the low-data regime, and at 256 px they help when mixed with few real patients. We release the code, the patient split, the pre-declared protocol and every run's configuration and metrics.

Reproducibility. Code, split file, protocol declarations with timestamps, result tables and figures: `https://github.com/wasiqnauman/SynthMRI` (branch `feat/publishable-pipeline`). One command reproduces the full experiment set on a single 16 GB GPU in about 40 hours; every reported number is traceable to a run directory that records the resolved configuration, git commit and metrics.

References

[1] Muhammad Usman Akbar, Wuhao Wang, and Anders Eklund. Beware of diffusion models for synthesizing medical images – a comparison with GANs in terms of memorizing brain MRI and chest x-ray images. *arXiv preprint arXiv:2305.07644*, 2023.

[2] Spyridon Bakas, Hamed Akbari, Aristeidis Sotiras, et al. Advancing the cancer genome atlas glioma MRI collections with expert segmentation labels and radiomic features. *Scientific Data*, 4:170117, 2017.

[3] Spyridon Bakas, Mauricio Reyes, Andras Jakab, et al. Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the BRATS challenge. *arXiv preprint arXiv:1811.02629*, 2018.

[4] Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. Demystifying MMD GANs. In *International Conference on Learning Representations (ICLR)*, 2018.

[5] M. Jorge Cardoso, Wenqi Li, Richard Brown, et al. MONAI: An open-source framework for deep learning in healthcare. *arXiv preprint arXiv:2211.02701*, 2022.

[6] Nicholas Carlini, Jamie Hayes, Milad Nasr, Matthew Jagielski, Vikash Sehwag, Florian Tramèr, Borja Balle, Daphne Ippolito, and Eric Wallace. Extracting training data from diffusion models. In *32nd USENIX Security Symposium*, pages 5253–5270, 2023.

[7] Richard J. Chen, Ming Y. Lu, Tiffany Y. Chen, Drew F. K. Williamson, and Faisal Mahmood. Synthetic data in machine learning for medicine and healthcare. *Nature Biomedical Engineering*, 5:493–497, 2021.

[8] Salman Ul Hassan Dar, Arman Ghanaat, Jannik Kahmann, Isabelle Ayx, Theano Papavassiliu, Stefan O. Schoenberg, and Sandy Engelhardt. Investigating data memorization in 3D latent diffusion models for medical image synthesis. In *Deep Generative Models (DGM4MICCAI), MICCAI Workshops*, volume 14533 of *LNCS*, pages 56–65, 2023.

[9] Virginia Fernandez, Walter Hugo Lopez Pinaya, Pedro Borges, Petru-Daniel Tudosiu, Mark S. Graham, Tom Vercauteren, and M. Jorge Cardoso. Can segmentation models be trained with fully synthetically generated data? In *Simulation and Synthesis in Medical Imaging (SASHIMI), MICCAI Workshops*, volume 13570 of *LNCS*, pages 79–90, 2022.

[10] Maayan Frid-Adar, Idit Diamant, Eyal Klang, Michal Amitai, Jacob Goldberger, and Hayit Greenspan. GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification. *Neurocomputing*, 321:321–331, 2018.

[11] Xiangming Gu, Chao Du, Tianyu Pang, Chongxuan Li, Min Lin, and Ye Wang. On memorization in diffusion models. *arXiv preprint arXiv:2310.02664*, 2023.

[12] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

[13] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[14] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 6840–6851, 2020.

[15] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18: 203–211, 2021.

[16] Tero Karras, Miika Aittala, Janne Hellsten, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Training generative adversarial networks with limited data. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 12104–12114, 2020.

[17] Amirhossein Kazerouni, Ehsan Khodapanah Aghdam, Moein Heidari, Reza Azad, Mohsen Fayyaz, Ilker Hacihaliloglu, and Dorit Merhof. Diffusion models in medical imaging: A comprehensive survey. *Medical Image Analysis*, 88:102846, 2023.

[18] Firas Khader, Gustav Müller-Franzes, Soroosh Tayebi Arasteh, et al. Denoising diffusion probabilistic models for 3D medical image generation. *Scientific Reports*, 13:7303, 2023.

[19] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[20] Xueyan Mei, Zelong Liu, Philip M. Robson, et al. RadImageNet: An open radiologic deep learning research dataset for effective transfer learning. *Radiology: Artificial Intelligence*, 4(5):e210315, 2022.

[21] Bjoern H. Menze, Andras Jakab, Stefan Bauer, et al. The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2015.

[22] Gustav Müller-Franzes, Jan Moritz Niehues, Firas Khader, et al. A multimodal comparison of latent denoising diffusion probabilistic models and generative adversarial networks for medical image synthesis. *Scientific Reports*, 13:12098, 2023.

[23] Walter H. L. Pinaya, Petru-Daniel Tudosiu, Jessica Dafflon, Pedro F. Da Costa, Virginia Fernandez, Parashkev Nachev, Sebastien Ourselin, and M. Jorge Cardoso. Brain imaging generation with latent diffusion models. In *Deep Generative Models (DGM4MICCAI), MICCAI Workshops*, volume 13609 of *LNCS*, pages 117–126, 2022.

[24] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022.

[25] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, volume 9351 of *LNCS*, pages 234–241, 2015.

[26] Gowthami Somepalli, Vasu Singla, Micah Goldblum, Jonas Geiping, and Tom Goldstein. Diffusion art or digital forgery? Investigating data replication in diffusion models. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6048–6058, 2023.

[27] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2021.

[28] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[29] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, 2018.